

12 + 13 Euclidean Geometry I

* Remember the proofs of theorems.

QUESTION 1

1.1 Complete the following statement:

The angle between the tangent and the chord is equal ...

(1)

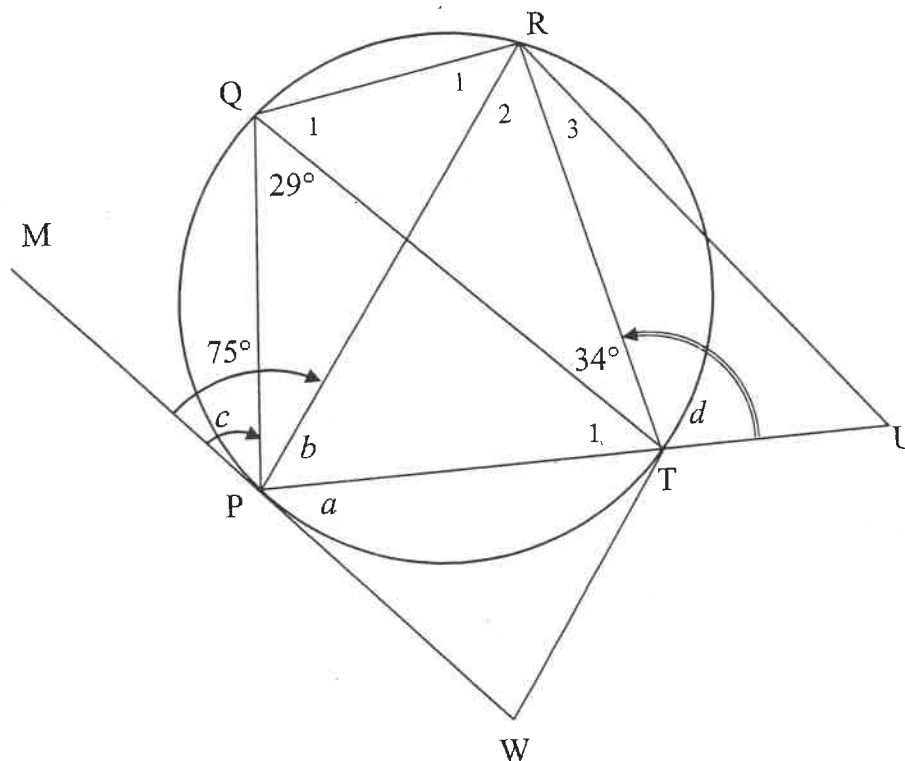
1.2 In the diagram points P, Q, R and T lie on the circumference of a circle. MW and TW are tangents to the circle at P and T respectively. PT is produced to meet RU at U.

$$\angle MPR = 75^\circ$$

$$\angle PQT = 29^\circ$$

$$\angle QTR = 34^\circ$$

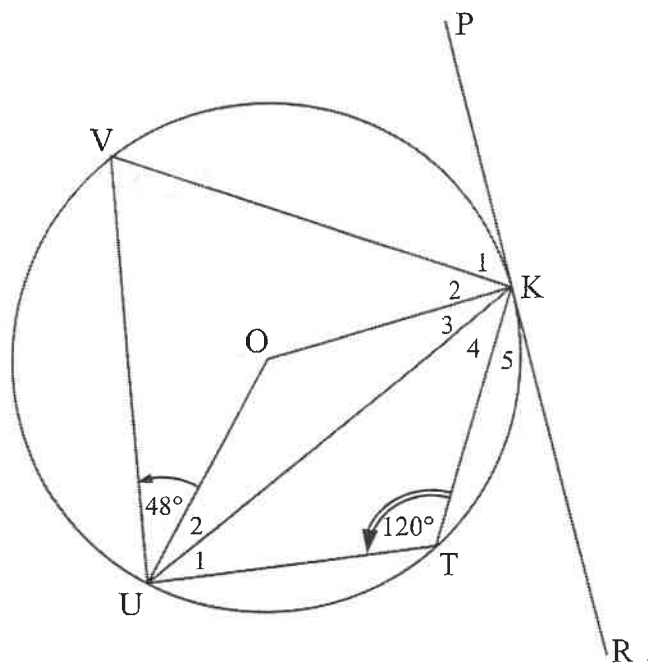
Let $\angle TPW = a$, $\angle RPT = b$, $\angle MPQ = c$ and $\angle RTU = d$, calculate the values of a , b , c and d .



(9)
[10]

QUESTION 2

In the diagram below, O is the centre of the circle KTUV. PKR is a tangent to the circle at K. $\angle UV = 48^\circ$ and $\angle KTU = 120^\circ$.



Calculate, with reasons, the sizes of the following angles:

- | | | |
|-----|--------------|------|
| 2.1 | $\hat{V}$ | (2) |
| 2.2 | $\angle KOU$ | (2) |
| 2.3 | $\hat{U}_2$ | (2) |
| 2.4 | $\hat{K}_1$ | (2) |
| 2.5 | $\hat{K}_2$ | (2) |
| | | [10] |

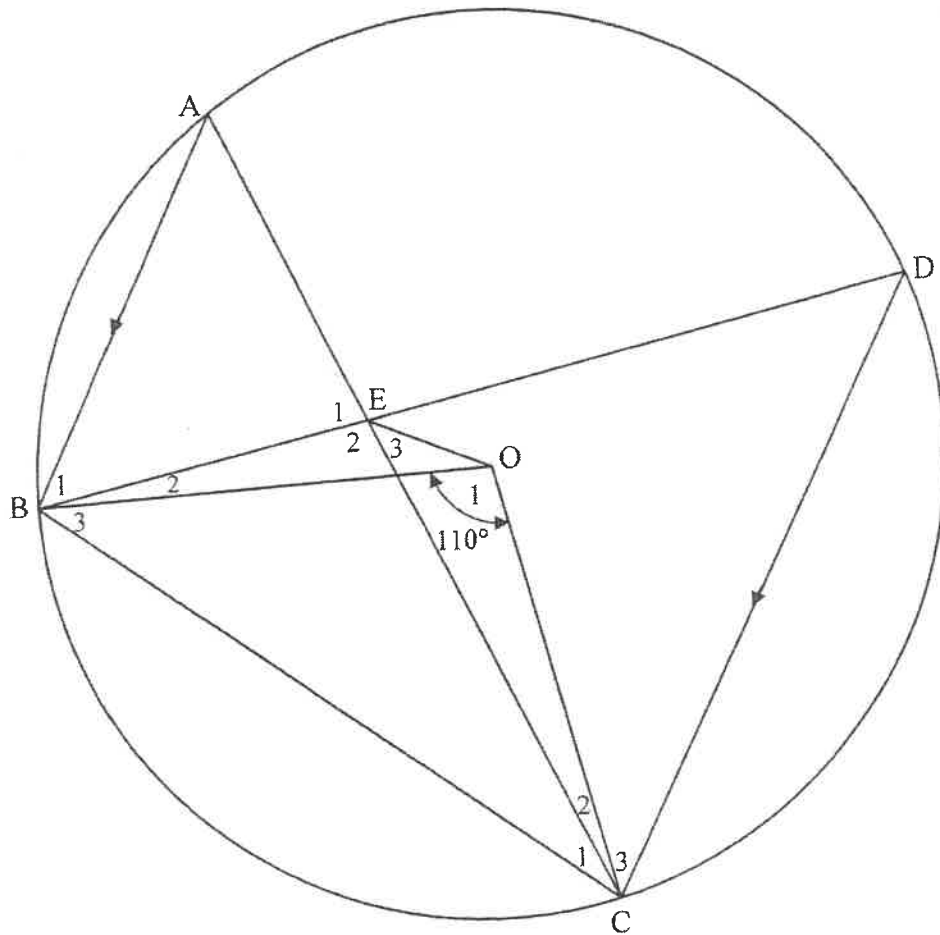
QUESTION 3

In the diagram below, the circle with centre O passes through A , B , C and D .

$AB \parallel DC$ and $\hat{BOC} = 110^\circ$.

The chords AC and BD intersect at E .

EO , BO , CO and BC are joined.



3.1 Calculate the size of the following angles, giving reasons for your answers:

3.1.1 $\hat{D}$

(2)

3.1.2 $\hat{A}$

(2)

3.1.3 $\hat{E}_2$

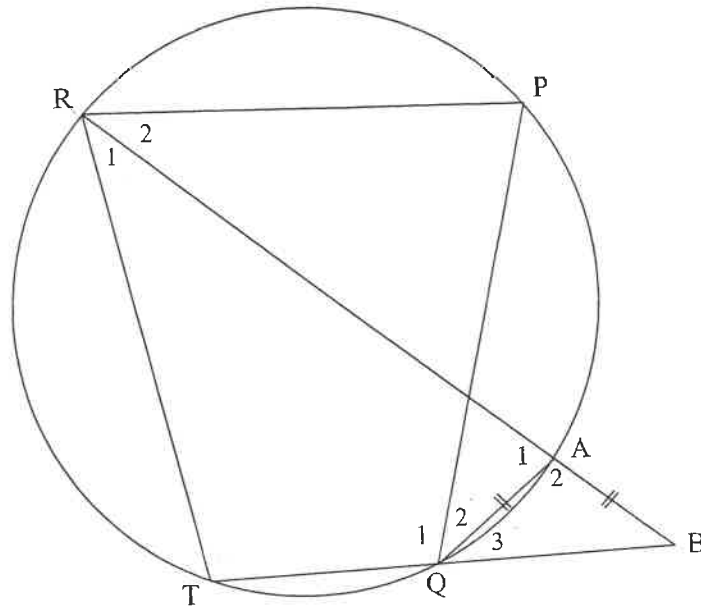
(4)

(2)

3.2 Prove that $BEOC$ is a cyclic quadrilateral.

QUESTION 4

In the diagram below, points R, P, A, Q and T lie on a circle. RA bisects $\hat{R}$ and $AB = AQ$. RA and TQ produced meet at B.



Prove that:

4.1 AQ bisects $\hat{PQB}$

(3)

4.2 $TR = TB$

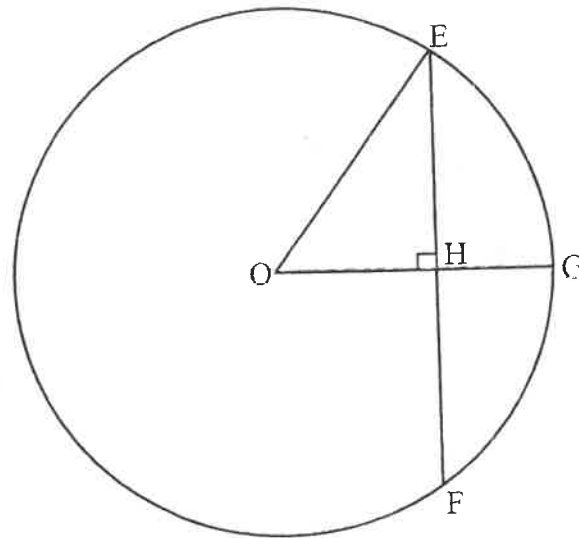
(2)

4.3 $\hat{P} = \hat{TRP}$

(3)
[8]

QUESTION 5

In the diagram below, EF is the chord of a circle with centre O. OG is perpendicular to EF.

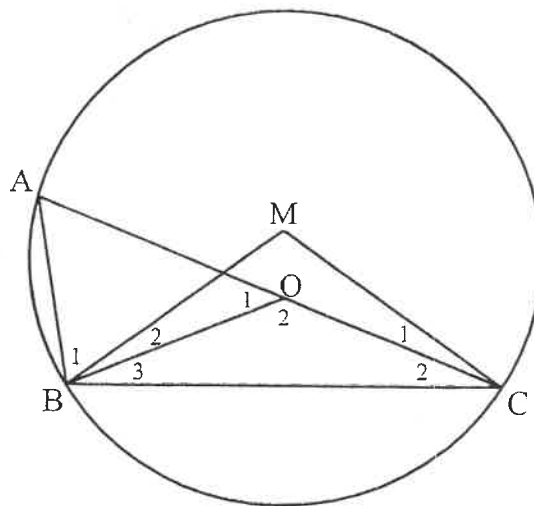


If $OE = x$ cm and $HG = 2$ cm, determine with reasons the size of EF (in its simplest form) in terms of x .

(4)

QUESTION 6

In the figure, M is the centre of the circle through A, B, and C. OB and AC bisect $\angle MBC$ and $\angle MCB$ respectively with O on AC. AB is joined.



Let $\hat{B}_2 = x$

6.1 Determine the size of $\hat{O}_2$ in terms of x .

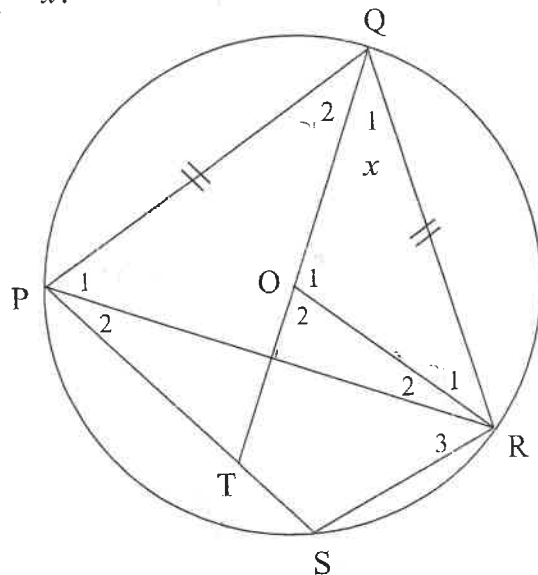
(4)

6.2 Prove that AO is a diameter of circle ABO.

(5)

QUESTION 7

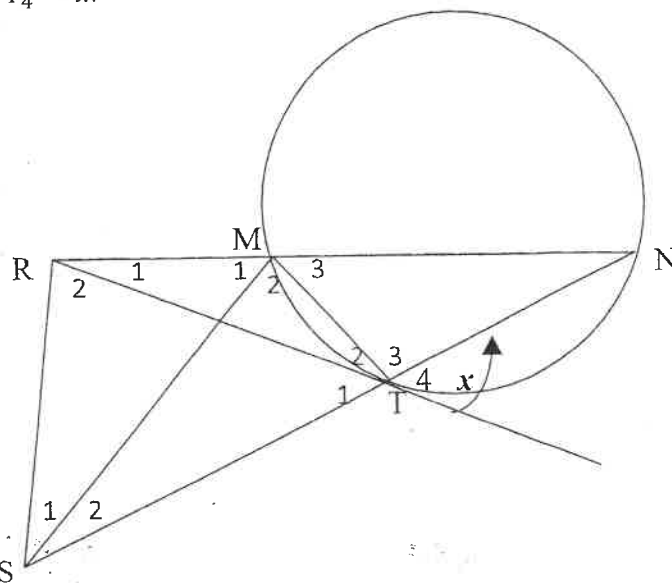
In the diagram below, O is the centre of the circle. P, Q, R and S are points on the circumference of the circle. TOQ is a straight line such that T lies on PS. $PQ = QR$ and $\hat{Q}_1 = x$.



- 7.1 Calculate, with reasons, $\hat{P}_1$ in terms of x . (3)
- 7.2 Show that TQ bisects $\hat{PQR}$. ($\hat{Q}_1 = \hat{Q}_2$) (3)
- 7.3 Show that STOR is a cyclic quadrilateral. (3)

QUESTION 8

RT is a tangent to the circle at T. NM and NT are produced to R and S respectively. $RS = RT$. Let $\hat{T}_4 = x$.

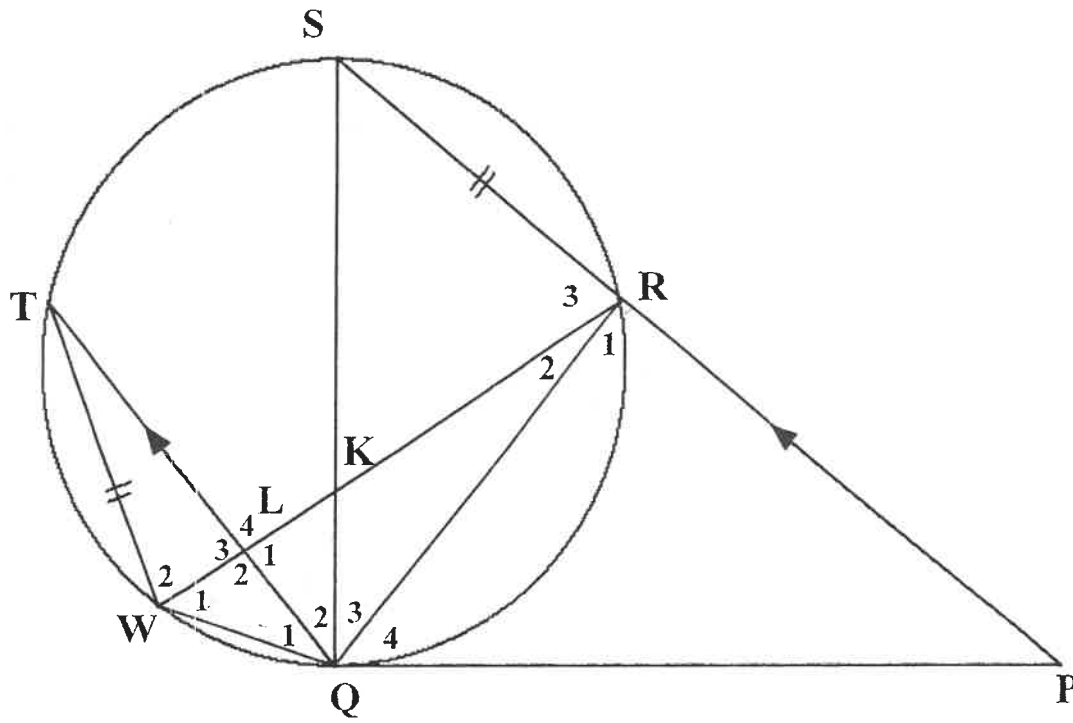


- 8.1 Write down, with reasons, three other angles that are equal to x . (5)
- 8.2 Prove that RSTM is a cyclic quadrilateral. (3)

QUESTION 9

In the diagram below, PQ is a tangent to circle SRQWT at Q.
 PRS is a straight line.
 RW cuts SQ and QT at K and L respectively.

$PS \parallel QT$, $RS = TW$ and $\hat{Q}_2 = x$.



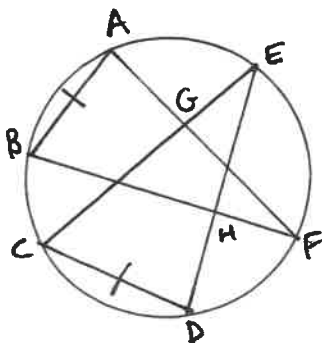
9.1 Find, with reasons, three other angles equal to x . (5)

Prove that :

9.2 $\hat{R}_1 = \hat{L}_3$ (5)

9.3 PRKQ is a cyclic quadrilateral. (6)

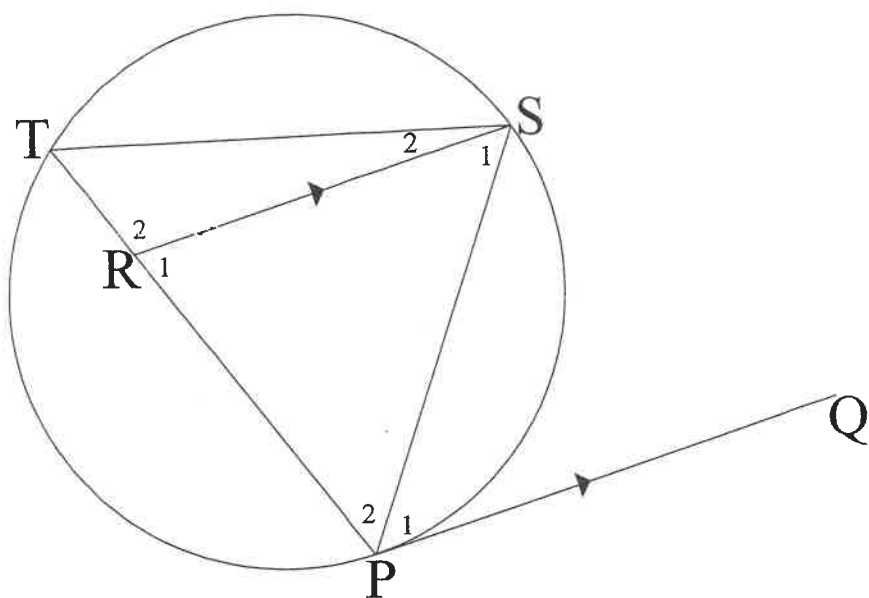
QUESTION 10



Prove that EFHG is a cyclic quadrilateral.

QUESTION 11

In the following sketch, PQ is a tangent to circle TSP at P. $RS \parallel PQ$.



Prove that PS is a tangent to circle TSR at S.

(5)

Euclidean Geometry 1: Solutions

Question 1:

1.1. is equal to the angle in the opposite segment, subtended by the chord.

1.2. $a = 29^\circ$ tan chord theorem
 $b = 180^\circ - 75^\circ - 29^\circ$ $\angle$'s on a straight line
 $b = 76^\circ$

$\hat{QPR} = 34^\circ$ $\angle$'s in same segment
 $\therefore c = 75^\circ - 34^\circ$
 $= 41^\circ$

$\hat{Q}_1 = 76^\circ$ $\angle$'s in same segment
 $d = \hat{Q} = 76^\circ + 29^\circ$ ext $\angle$ of cyclic quad.
 $= 105^\circ$

Question 2:

2.1 $\hat{V} = 180^\circ - 120^\circ$ opp $\angle$'s of cyclic quad.
 $= 60^\circ$

2.2 $\hat{KOU} = 2\hat{V}$ $\angle$ at centre = $2 \times \angle$ at circ.
 $= 2(60^\circ)$
 $= 120^\circ$

2.3 $\hat{K}_3 = \hat{U}_2$ $\angle$'s opp equal radii
 $\therefore 2\hat{U}_2 = 180^\circ - \hat{O}_2$ int $\angle$'s of Δ
 $\hat{U}_2 = 30^\circ$

(2)

$$\begin{aligned}
 2.4 \quad \hat{K}_1 &= 48^\circ + \hat{U}_2 && \text{tan chord theorem} \\
 &= 48^\circ + 30^\circ \\
 &= \underline{78^\circ}
 \end{aligned}$$

$$\begin{aligned}
 2.5 \quad \hat{K}_1 + \hat{K}_2 + \hat{K}_3 &= 120^\circ && \text{tan chord theorem} \\
 \hat{K}_2 &= 120^\circ - 78^\circ - 30^\circ \\
 \hat{K}_2 &= \underline{12^\circ}
 \end{aligned}$$

Question 3

$$\begin{aligned}
 3.1.1 \quad \hat{D} &= \frac{1}{2} \hat{O}, && \angle \text{ at centre} = 2 \times \angle \text{ at circ} \\
 &= \frac{1}{2} (110^\circ) \\
 &= \underline{55^\circ}
 \end{aligned}$$

$$3.1.2 \quad \hat{A} = \hat{D} = 55^\circ \quad \angle\text{'s in same segment}$$

$$\begin{aligned}
 3.1.3 \quad \hat{B}_1 &= \hat{D} = 55^\circ && \text{alt } \angle\text{'s } AB \parallel CD \\
 \hat{E}_2 &= \hat{A} + \hat{B}_1, && \text{ext } \angle \text{ of } \triangle \\
 &= \underline{110^\circ}
 \end{aligned}$$

$$3.2 \quad \hat{E}_2 = \hat{O}_1 = 110^\circ \quad \text{proven}$$

$\therefore$ BEOC is a cyclic quad
line subtends equal $\angle$'s

Question 4

4.1. $\hat{Q}_2 = \hat{R}_2$

$$\hat{Q}_3 = \hat{R}_1$$

but $\hat{R}_1 = \hat{R}_2$

$$\therefore \hat{Q}_2 = \hat{Q}_3$$

$\angle$'s in same segment
ext $\angle$ of cyclic quad
gives

4.2. $\hat{Q}_3 = \hat{R}_1$

$$\hat{Q}_3 = \hat{B}$$

$$\therefore \hat{B} = \hat{R}_1$$

$$\therefore \angle R = \angle B$$

proven

$\angle$'s opp equal sides

sides opp equal $\angle$'s

4.3. $\hat{A}_1 = \hat{Q}_3 + \hat{B}$

ext $\angle$ of Δ

but $\hat{Q}_3 = \hat{R}_1 = \hat{R}_2$ proven
 $\hat{B} = \hat{Q}_3$

$$\therefore \hat{A}_1 = \hat{R}_1 + \hat{R}_2$$

$$\hat{A}_1 = \hat{P}$$

$\angle$'s in same segment

$$\therefore \hat{P} = \hat{R}_1 + \hat{R}_2$$

Question 5:

$$OE = OG = x \quad \text{radii}$$

$$\therefore OH = x - 2$$

$$EH^2 = OE^2 - OH^2 \quad \text{Pythag}$$

$$= x^2 - (x-2)^2$$

$$= x^2 - (x^2 - 4x + 4)$$

$$EH^2 = 4x - 4$$

$$EH = \sqrt{4x-4}$$

$$= 2\sqrt{x-1}$$

$$EH = HF$$

line from centre $\perp$ chord

$$\therefore EF = 2EH$$

$$= 4\sqrt{x-1}$$

Question 6:

$$6.1 \quad \hat{B}_2 + \hat{B}_3 = x \quad \text{given}$$

$$\hat{B}_2 + \hat{B}_3 = \hat{C}_1 + \hat{C}_2 = 2x \quad \angle\text{'s opp equal radii}$$

$$\text{but } \hat{C}_1 = \hat{C}_2$$

given

$$\therefore \hat{C}_2 = x$$

$$\therefore \hat{O}_2 = 180^\circ - \hat{B}_3 - \hat{C}_2 \quad \text{int } \angle\text{'s of } \Delta$$

$$= 180^\circ - 2x$$

(5)

$$6.2, \quad \hat{M} = 180^\circ - (\hat{B}_2 + \hat{B}_3) - (\hat{C}_1 + \hat{C}_2) \text{ int } \angle \text{'s of } \Delta$$

$$= 180^\circ - 4x$$

$$\hat{A} = \frac{1}{2} \hat{M} \quad \angle \text{ at centre} = 2 \times \angle \text{ at circ}$$

$$\hat{A} = \frac{1}{2} (180^\circ - 4x)$$

$$\hat{A} = 90^\circ - 2x$$

$$\hat{O}_2 = \hat{A} + \hat{B}_1 + \hat{B}_2 \quad \text{ext } \angle \text{ of } \Delta$$

$$180^\circ - 2x = 90^\circ - 2x + \hat{B}_1 + \hat{B}_2$$

$$90^\circ = \hat{B}_1 + \hat{B}_2$$

$\therefore AO$ is a diameter line subtends 90°

Question 7

$$7.1, \quad \hat{R}_1 = x \quad \angle \text{'s opp equal radii}$$

$$\hat{O}_1 = 180^\circ - 2x \quad \text{int } \angle \text{'s of triangle}$$

$$\hat{P}_1 = \frac{1}{2} (\hat{O}_1) \quad \angle \text{ at centre} = 2 \times \angle \text{ at circ}$$

$$= \frac{1}{2} (180^\circ - 2x)$$

$$= 90^\circ - x$$

$$7.2, \quad \hat{R}_1 + \hat{R}_2 = 90^\circ - x \quad \angle \text{'s opp equal sides}$$

$$\hat{Q}_2 + \hat{Q}_1 + \hat{R}_1 + \hat{R}_2 + \hat{P}_1 = 180^\circ \text{ int } \angle \text{'s of } \Delta$$

$$\hat{Q}_2 = 180^\circ - x - (90^\circ - x) - (90^\circ - x)$$

$$= x$$

$$\therefore \hat{Q}_1 = \hat{Q}_2$$

(6)

7.3. $\hat{S} = 180^\circ - \hat{Q}$
 $= 180^\circ - 2x$

opp $\angle$'s of cyclic quad

$$\hat{O}_2 = \hat{Q}_1 + \hat{R}_1$$

$$= 2x$$

ext $\angle$ of Δ

$$\hat{S} + \hat{O}_2 = 180^\circ - 2x + 2x$$

$$= 180^\circ$$

$\therefore$ STOR is a cyclic quad opp $\angle$'s of quad
 supplementary

Question 8:

8.1. $\hat{T}_1 = x$

vert opp $\angle$'s

$$\hat{M}_3 = x$$

tan chord theorem

$$\hat{S}_1 + \hat{S}_2 = x$$

 $\angle$'s opp equal sides

8.2. $\hat{M}_3 = \hat{S}_1 + \hat{S}_2 = x$

proven

$\therefore$ RSTM is a cyclic quad ext $\angle =$ opp int $\angle$

Question 9

9.1. $\hat{Q}_2 = x$

give's

$$\hat{Q}_2 = \hat{S} = x$$

alt $\angle$'s $TQ \parallel SR$

$$\hat{Q}_4 = \hat{S} = x$$

tan chord theorem

$$\hat{W}_1 = \hat{S} = x$$

 $\angle$'s in same segment

9.2. $\hat{R}_1 = \hat{Q}_2 + \hat{Q}_3$
 $\therefore \hat{R}_1 = x + \hat{Q}_3$

alt $\angle$'s $PS \parallel TQ$

$\hat{L}_3 = \hat{W}_1 + \hat{Q}_1$ ext $\angle$ of $\triangle$
 $\therefore \hat{L}_3 = x + \hat{Q}_1$

but $\hat{Q}_1 = \hat{Q}_3$ equal chords equal $\angle$'s

$\therefore \hat{L}_3 = x + \hat{Q}_3$

$\therefore \hat{R}_1 = \hat{L}_3$

9.3. $\hat{Q}_3 + \hat{Q}_4$
 $= x + \hat{Q}_3$
 $= \hat{R}_1$

proven

but $\hat{R}_1 = \hat{L}_3$

$\hat{L}_3 = \hat{L}_1$

$\hat{L}_1 = \hat{R}_3$

$\therefore \hat{Q}_3 + \hat{Q}_4 = \hat{R}_3$

proven

vert opp $\angle$'s

alt $\angle$'s $TQ \parallel PS$

$\therefore PRKQ$ is a cyclic quad ext $\angle$ of quad
 $=$ opp int $\angle$.

Question 10

$\hat{E} = \hat{F}$

equal chords; equal $\angle$'s

$\therefore EFHG$ is a cyclic quad line subtends equal $\angle$'s.

Question 11

$$\hat{P}_1 = \hat{T}$$

$$\hat{P}_1 = \hat{S}_1$$

$$\therefore \hat{T} = \hat{S}_1$$

tan chord theorem

alt $\angle$'s RS $\parallel$ PQ

$\therefore$ PS is a tangent converse tan chord theorem